FIELD GUIDE

Rooftop Solar: Where the Power Goes

Generation losses, cleaning, maintenance and measurement, from a 930 kW rooftop plant

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930 kW	70% → 87%	27
rooftop plant	reported PV performance, before and after the work	loss mechanisms in the register

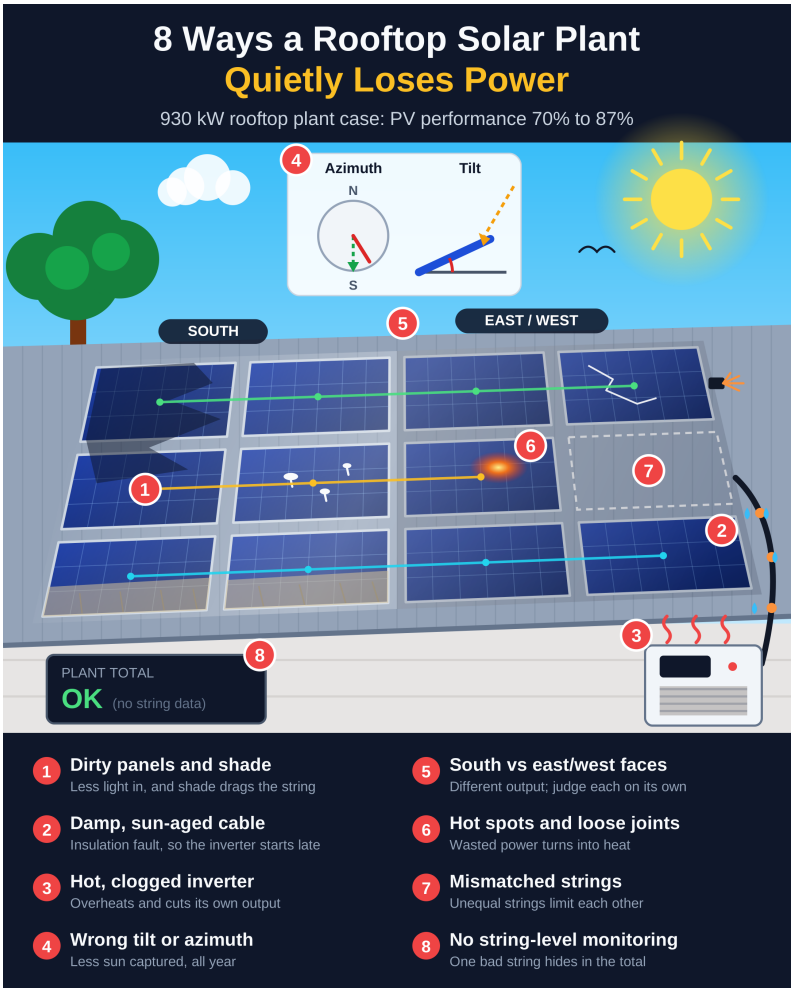
How to read the numbers. The 70% and 87% figures are the plant's reported performance before and after the work. In formal reporting the metric should be named (for example performance ratio), and both periods compared under similar sun, temperature and availability. The operating figures in this guide are as reported for the plant and have not been independently audited.

Inside. What I found, a 27-item loss register, cleaning and water, a maintenance schedule, checking the numbers, an audit workflow, and notes on DG–PV and carbon.

What I found

A solar plant rarely fails in a way you can see. Nothing trips and no alarm goes off. The generation number is just lower than it should be. On this 930 kW rooftop plant most of what I found was ordinary: dirt, an aged cable, a hot inverter, a tilt nobody had checked. None of it was dramatic. Added together it was the gap between about 70% and 87%.

The picture shows the eight places I look first. The register on the next pages goes through all 27 in detail.



Illustration, not a photo of the plant.

The loss register: 27 things to check

Each line is something to check for, not proof that it exists on a particular plant. Confirm at site before you count it.

Modules, DC side and strings (01 to 14)

#	What goes wrong	What I check
01	**Dust and dirt on the glass.** Dust, pollen and fallout cut the light reaching the cells.	Map dirt by roof zone. Compare a clean panel with a dirty one. Match against the cleaning log and rainfall.
02	**Hard bird droppings.** Opaque patches shade cells and can create hot areas.	Look for hot spots under deposits. Remove with the module maker's approved, non-abrasive method.
03	**Mud along the lower frame edge.** Shades the bottom cells and holds moisture.	Check the lower frame lip and drainage. Note zones where it keeps coming back.
04	**Shade from nearby objects.** Ducts, tanks, parapets, trees and new structures throw moving shadows.	Do a time-of-day shade survey. Compare with string layout and MPPT assignment.
05	**Light-induced degradation (LID).** Output settles lower after the first exposure to sun.	Compare against module type, warranty tolerance and initial flash data.
06	**Potential-induced degradation (PID).** Voltage, humidity and heat can drive leakage and loss.	Check string voltage, insulation and grounding. Look at the pattern of degradation across modules.
07	**Micro-cracks and snail trails.** Cracks from handling, foot traffic or vibration.	Keep people off the modules. Use EL testing when it is justified and compare with string output.
08	**Hot spots and failed bypass diodes.** Damaged or shaded cells heat up. A failed diode switches off part of a module.	IR scan in good sun and under load. Check the diode and module with the maker's procedure.
09	**EVA yellowing and delamination.** Ageing encapsulant lets moisture in and speeds corrosion.	Look for bubbles, colour change and edge lifting. Note the warranty position.
10	**Insulation resistance drop in the monsoon.** Aged or damaged DC insulation leaks when wet, and the inverter can start late.	Test before the monsoon with an approved insulation resistance method, isolated, by qualified people.
11	**Loose or mismatched MC4 connectors.** Poor crimps or cross-mated types heat up and can arc.	Confirm the same make and type. Inspect the crimp and look at the thermal image.
12	**String current mismatch.** Uneven dirt, module mismatch or ageing pulls one string down.	Trend string currents in steady sun. Compare strings with the same orientation.
13	**DC cable losses (I^2R).** Long or undersized cable wastes power as heat.	Check conductor size, route length, joints and voltage drop against the design.
14	**Blown fuses and open isolators.** An open string can hide inside the inverter total.	Use string monitoring or a safe current check. Inspect fuses and isolators.

Inverter, AC, design and grid (15 to 27)

#	What goes wrong	What I check
15	**Corroded AJB or DCDB busbars.** Moisture corrodes terminals and raises resistance.	Check enclosure seals, glands and condensation. Check terminal temperature.
16	**High earth resistance or a failed SPD.** Weak earthing or a dead surge protector reduces protection and can cause downtime.	Test the earth system. Inspect SPD status against the design and the maker's guidance.
17	**Inverter thermal derating.** Dusty heatsinks, failed fans or high ambient heat cut the output.	Trend inverter temperature, fan state, ambient conditions and the derating flag.

#	What goes wrong	What I check
18	**DC-to-AC clipping.** Output is capped when DC power exceeds the inverter's AC limit.	Compare clipping time with the design DC/AC ratio.
19	**MPPT mismatch.** Different roof orientations or string types on one MPPT lose harvest.	Review string-to-MPPT mapping, voltage windows and orientation grouping.
20	**AC over-voltage trips.** High feeder voltage disconnects inverters at high generation and low load.	Trend voltage at the inverter terminals and the PCC. Review protection events.
21	**Zero-export or DG sync hunting.** Controller or reverse-power settings hold PV back as load or DG output changes.	Trend load, PV setpoint, DG kW, frequency, the reverse-power relay and controller mode.
22	**AC panel and transformer losses.** Loose terminations, cable losses and transformer losses.	Safe thermal checks of terminations. Review cable sizing, loading and transformer temperature.
23	**Sub-optimal tilt.** Tilt affects yearly sun capture, drainage and self-cleaning.	Compare installed tilt with the design yield model and roof limits.
24	**Azimuth and multi-face mismatch.** South, east, west and north faces have different generation curves.	Benchmark each roof face on its own. Allocate strings and MPPTs to match.
25	**Inter-row self-shading.** Tight row spacing shades panels when the sun is low.	Do a seasonal shading check and confirm the real shadow windows.
26	**Module temperature.** Hot cells lose power compared with rated conditions.	Trend module temperature against irradiance. Check rear ventilation and mounting gap.
27	**Reactive power and power factor.** Reactive demand can limit active power and affect billing.	Trend PF and kvar. Check the inverter capability curve and PCC measurements.

Losses that are not faults

No list is complete. The register above is for things you can find and fix. These losses are real too, but different in kind, so keep them separate and don't chase them as faults. I did not see every item on this plant.

Loss	Why it is separate	What I do
Grid outages and power failures	The plant is healthy but can't run or export.	Log outage hours separately and keep them out of the performance figure.
Curtailment and export limits	A control setting, not a defect.	Log the limited hours. Check controller settings.
Less sun: cloud, haze, smoke, monsoon	The resource itself.	Compare with measured plane-of-array irradiance, not calendar months.
Built-in and design losses: temperature, reflection, low light, module tolerance, yearly degradation, inverter efficiency, standby use, transformer and cable losses	Part of every plant.	Put them in the expected-output model so they aren't mistaken for faults. Act only if they exceed design.
Site damage: nests under panels, animal damage to cables, lightning or surge damage, loose clamps in wind, weeds	Happens around the plant, outside the electrical design.	Walk the roof each season and after storms. Photograph and log.
Slow repair: no spares, slow response	Turns a small fault into days of lost generation.	Keep critical spares. Agree response times.
Bad measurement: sensor drift, wrong irradiance data, meter scaling, clock errors	The plant is fine and the number is wrong.	See the measurement mistakes I look for.

Cleaning without causing new problems

Cleaning is maintenance, not a favour to the panels. Too rarely and you leave loss on the glass that you could have recovered. Done badly it scratches coatings, leaves mineral deposits or pushes water where it shouldn't go. On this plant the panels were cleaned about twice a month. The right interval is whatever your site's dirt rate says, so check energy before and after instead of copying a schedule.

Water

Water	What can happen	What to do
High TDS or hard water	Mineral spots or a scale film after it dries. Cuts light and holds dust.	Test the source water. Follow the module maker's water spec. Consider filtration or demineralised water where it is justified.
Grit or suspended solids	Scratches the glass when you brush or wipe.	Filter or settle the water. Keep brushes and cloths clean. Never drag grit across the glass.
Wrong pH or contamination	Can stain or attack coatings, frames, seals and backsheets.	Don't assume borewell, tanker or recycled water is suitable. Check it.
Hot module, cold water	Thermal shock, and fast drying that leaves deposits.	Clean in cooler hours and follow the maker's temperature guidance.

TDS alone doesn't describe water quality. Conductivity, hardness, pH and suspended solids matter too.

Chemicals: only when justified

Chemical cleaning is for oily fallout, soot or sticky residue that water won't shift. It is not an upgrade. Frequent use or the wrong strength can damage the glass coating, frame, seals or backsheet.

- Identify the dirt before you choose water or chemicals.
- Follow the module maker's cleaning manual and approved chemical list.
- Record the product, dilution, contact time, rinse and approval reference.
- Never improvise mixtures of acids, alkalis, solvents or detergents.
- Use soft, non-abrasive tools. No blades, scouring pads or dry scraping.
- Keep jets away from connectors, junction boxes, cable glands and module edges.
- If compatibility is unknown, don't start. Run a controlled trial with engineering review first.

What I do next

What I see	What I do
Soiling loss is rising	Plan a clean and compare normalised energy before and after.
Oily deposits stay after a water wash	Identify the dirt. Trial an approved chemical on a small area.
White spots after cleaning	Test TDS, conductivity and hardness. Review the rinse and drying.
No recovery after cleaning	Don't just clean more often. Check shading, strings, sensors, inverter limits and grid restrictions.
Scratches or coating damage	Stop, document it, and change tools and method with the maker's advice.

Maintenance schedule

The intervals below are a starting point. Final intervals follow the maker's instructions, site risk, warranty and the applicable standards.

Activity	When	What I keep
Module cleaning	Start with a site schedule, then adjust to the soiling rate and the measured response.	Cleaning log, water test, photos, before and after energy.
IR thermography	Once a year to start, or when there is a reason. After faults or repairs. Needs good sun and load.	Thermal image, load, irradiance, ambient temperature, defect ID.
String current and voltage	Monthly trend. Chase anomalies quickly.	String comparison, irradiance context, action record.
String insulation resistance	Before the monsoon, and after suspected damage or water ingress. Qualified people only.	Test voltage, instrument ID, isolation boundary, conditions, result.
DC connectors, AJB, DCDB	Routine visual check. Thermal inspection when condition suggests it.	Connector make and type, crimp details, enclosure condition, thermal finding.
Inverter cooling	Routine alarm review and the maker's service interval.	Fan and filter status, heatsink temperature, derating events, service log.
AC panels and transformer	Scheduled, with safe thermal and electrical checks.	Load at the time, thermal images, torque records where permitted, test reports.
DG–PV controller and protection	After any change, and in periodic approved maintenance.	Setpoints, event logs, interlock and protection test records.
Sensors and meters	On the calibration schedule, and whenever readings look wrong.	Calibration certificate, ratio, scaling, configuration backup.

Safety first. PV DC circuits can stay live in daylight. Isolation, lockout and tagout, arc-flash precautions, work-at-height controls, roof load limits and the maker's instructions are not optional. Only qualified people should do insulation resistance tests, open enclosures, inspect live equipment or change DG–PV protection settings. This guide doesn't replace a method statement, a risk assessment or a statutory requirement.

Thermography: what I found

I tested it. It was done when the plant was installed, then not again for years. Now I do it about once a year, or when there is a reason. On this plant it hasn't added much so far. The faults that cost generation were found faster by the daily string data, insulation resistance tests and plain inspection. I still keep it as a periodic check, because it is a quick way to look for hot connectors and hot cells without touching anything.

If you do it, it only means something when you record the conditions: irradiance, ambient temperature, equipment load or current, wind and the time. Compare similar parts under similar conditions. Keep to safe approach distances, and don't open a live enclosure just to get a picture.

Check the numbers before you trust them

A single “efficiency” figure can hide very different causes, and it misleads if nobody has defined what it is divided by. Name the metric first.

Metric	Formula	What it tells you
Specific yield	AC energy (kWh) ÷ installed DC capacity (kWp)	Output per installed kWp, comparable across periods and sites.
Performance ratio (PR)	Final yield ÷ reference yield × 100	Plant performance separated from the sun you happened to get, with sensors and meter boundary defined.
Capacity utilisation factor	Actual energy ÷ (rated capacity × hours in period) × 100	How much of the installed capacity was used over the period.
Availability	Available time ÷ eligible scheduled time × 100	Time the equipment could run. Decide how grid outage and planned shutdowns are treated.

Compare like with like

Compare periods with similar plane-of-array irradiance, module temperature, availability and metering boundary. Keep “energy shortfall” (a gap) separate from “recoverable opportunity” (what you can actually get back).

Measurement mistakes I look for

Mistake	How I check
Wrong CT/PT ratio or meter multiplier	Compare SCADA with the meter display, drawings and approved ratio settings.
Wrong register address or word order	Verify the Modbus map: offset, data type, sign, byte and word order, scale factor.
Wrong units	Check W/kW, Wh/kWh, kWh/MWh and DC/AC capacity labels end to end.
Clocks or intervals out of step	Sync inverter, PLC, gateway, meter and server clocks. Align interval boundaries.
Missing or stale data shown as zero or live	Flag gaps and stale values. Don't turn missing data into zero or pass old values off as live.
Energy counted twice	Use counter delta logic and handle resets and rollovers.
Import and export swapped	Check CT polarity, meter direction and the sign convention.
Wrong denominator in PR	Check DC capacity, plane-of-array irradiance, reference irradiance and the meter boundary.
Sensor drift or poor placement	Check cleanliness, orientation, calibration, and that it sits in the array plane.
Cached dashboard value	Show a timestamp and quality status. Label cached values separately from live ones.

Audit workflow and priorities

A repeatable order of work means every finding can be traced to a measurement, an action and a result.

Step	What to do	What you end up with
1. Set the boundary	Confirm plant capacity, meter boundary, reporting period, grid, PV and DG sources, and time sync.	A signed measurement boundary and asset register.
2. Baseline performance	Collect energy, irradiance, temperature, availability, alarms, grid outage and DG fuel data.	A baseline KPI pack and a data-quality review.
3. Walk the assets	Modules, strings, connectors, DCDB, inverter, AC panel, transformer, roof structure and drainage.	A photo-backed fault register with locations and severity.
4. Test the priority faults	Approved IR, string current, insulation resistance, earthing, meter and control tests as relevant.	Test records with method, conditions and instrument details.
5. Size the opportunity	Estimate energy or fuel impact with a clear baseline and a stated uncertainty.	A ranked loss register, with no double counting.
6. Fix and verify	Do the approved work, then compare matched data after.	Closure evidence and a verified benefit.

Priorities

Priority	Typical example	Response
P1 Safety or fire risk	Arcing connector, exposed live conductor, failed protection, unsafe roof access.	Make safe immediately under site procedure, with a qualified electrical response.
P2 High energy or reliability	Open string, inverter derating, repeated grid trip, significant hot spot.	Investigate promptly and estimate the energy and reliability impact.
P3 Performance	Soiling, orientation mismatch, sensor drift, moderate cable loss.	Plan the fix and verify the normalised improvement.
P4 Optimisation and data hygiene	Dashboard scaling, reporting logic, documentation gaps.	Fix the calculation or configuration and protect the history.

Give every finding an ID, a date and time, the asset location, the symptom, the evidence, the suspected cause, a priority, how the impact was estimated, an owner, a due date, the action taken and a check afterwards. Don't close a fault because the alarm went away. Close it when you have confirmed the condition is fixed.

What the maintenance contract costs against what it needs to return

The annual maintenance contract costs about 7 lakh rupees. At 15 to 17 rupees a unit, it pays for itself if it recovers about 41,000 to 47,000 units a year. That is roughly 3% of what this plant generates (13 to 16 lakh units a year), or 115 to 130 units a day.

My estimate of the units recovered is well above that, but it rests on the before and after percentages sharing one baseline, so I treat it as an estimate until I can replace it with actual before and after generation. The tariff is my all-in figure including demand charges. Solar units don't reliably reduce maximum demand, so the part of the bill that falls with each unit can be lower.

DG–PV and carbon: what the numbers do and don't prove

Reported figures

Item	Value	Note
Generator	630 kVA diesel	Nameplate rating.
Fuel rate before DG–PV	about 160 L/h	Reported.
Fuel rate after DG–PV	about 90–100 L/h	Reported.
Running hours	100 hours a month assumed, both periods	Depends on grid outages.
Fuel per month	16,000 L before, 9,000–10,000 L after	160 × 100, and (90 to 100) × 100.
Fuel avoided	6,000–7,000 L a month (37.5–43.75% lower rate)	Provisional.

Why provisional. A generator burns less per hour if it is doing less work. To call this a saving I need DG kWh and litres per kWh, for comparable loading, run hours, outage conditions and engine condition. The format, with made-up numbers: 9,500 L for 42,000 kWh is 0.226 L/kWh.

When PV and DG share a bus

- Keep to the engine maker's minimum loading to avoid wet-stacking.
- Verify reverse-power protection, so the generator isn't motored.
- Test load-step response, PV ramp limits and frequency stability, including sudden load rejection (ANSI 32) and fast drops in irradiance.
- Trend DG kW, kVA, PF, frequency, PV kW, plant load and controller setpoints together, at a fast sample rate.
- Don't lengthen relay delays or disable protection to hide nuisance trips.

Carbon

With an indicative diesel factor of 2.6853 kg CO_2e per litre, 6,000–7,000 L a month avoided is about 16.1–18.8 tonnes of direct diesel emissions a month (roughly 193 to 226 tonnes if the pattern held for a year). Confirm the factor and method for the standard you report under. Solar energy that displaces grid power is a different calculation, using the grid factor for the reporting year. Don't apply a diesel factor and a grid factor to the same kWh. A claim like “x% lower plant carbon” needs a baseline and a post-project inventory on the same boundary.

For scale: 100,000 kWh a month of solar against 450,000 kWh of reported plant use is a 22.2% solar share. That is a share of energy, not a performance ratio.

About this guide. I wrote it from my own work on a 930 kW rooftop plant. The operating figures are as reported for the plant and have not been independently audited. It is not a certified energy audit or carbon inventory, and a line on the register is something to check for, not proof it exists on your roof.

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